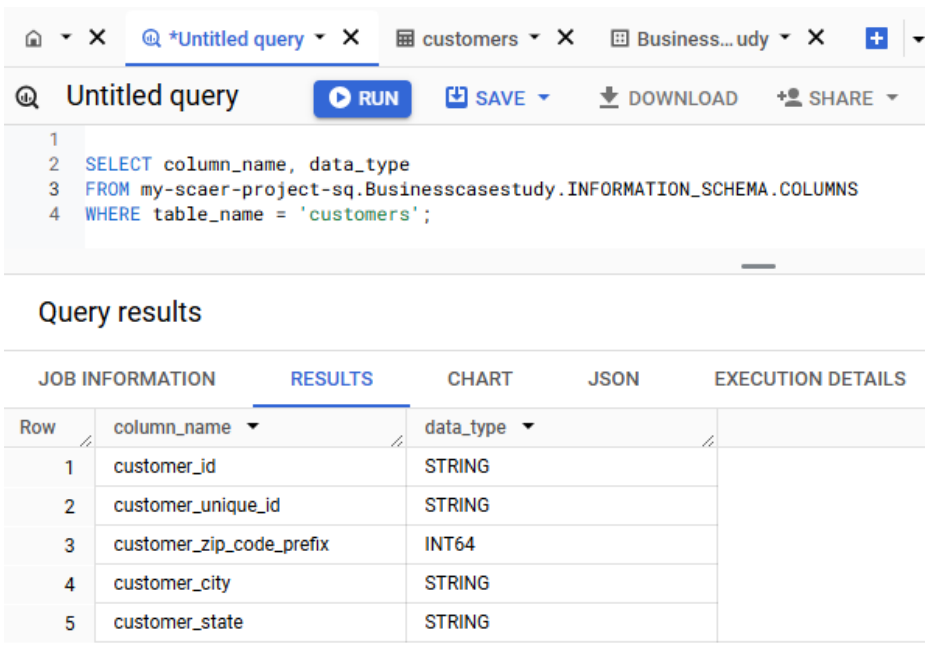
Q1: **Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:**

1. Data type of all columns in the "customers" table.

Query Ans:

```
SELECT column_name, data_type
FROM my-scaer-project-sq.Businesscasestudy.INFORMATION_SCHEMA.COLUMNS
WHERE table_name = 'customers';
```

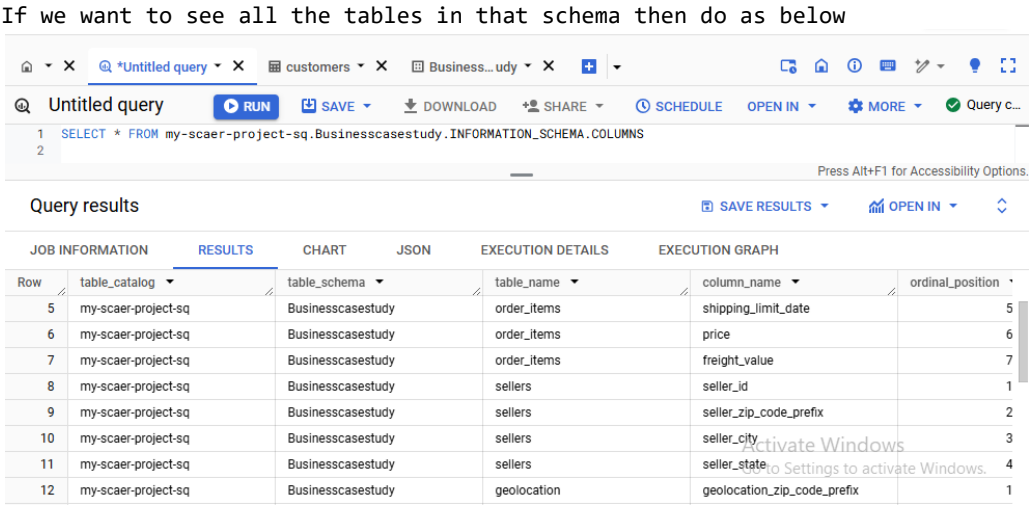
Output screenshot of 10 rows:



Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

Insights:

- Catalog name is nothing but Database name. In our case database name is ‘my-scaer-project-sq’.
- Schema name is ‘Businesscasestudy’
- If we want to see all the tables in that schema then do as below



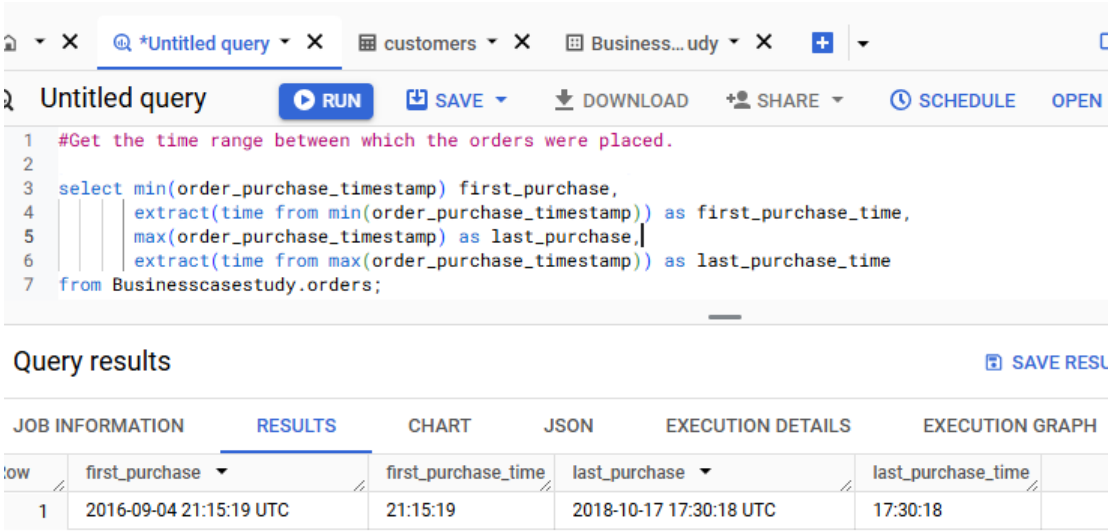
Row	table_catalog	table_schema	table_name	column_name	ordinal_position
5	my-scaer-project-sq	Businesscasestudy	order_items	shipping_limit_date	5
6	my-scaer-project-sq	Businesscasestudy	order_items	price	6
7	my-scaer-project-sq	Businesscasestudy	order_items	freight_value	7
8	my-scaer-project-sq	Businesscasestudy	sellers	seller_id	1
9	my-scaer-project-sq	Businesscasestudy	sellers	seller_zip_code_prefix	2
10	my-scaer-project-sq	Businesscasestudy	sellers	seller_city	3
11	my-scaer-project-sq	Businesscasestudy	sellers	seller_state	4
12	my-scaer-project-sq	Businesscasestudy	geolocation	geolocation_zip_code_prefix	1

2. Get the time range between which the orders were placed.

Query Ans:

```
select min(order_purchase_timestamp) first_purchase,
       extract(time from min(order_purchase_timestamp)) as first_purchase_time,
       max(order_purchase_timestamp) as last_purchase,
       extract(time from max(order_purchase_timestamp)) as last_purchase_time
from Businesscasestudy.orders;
```

Output screenshot of 10 rows: Our desired result is time range i.e.,first_purchase_time and last_purchase_time.



Insights:

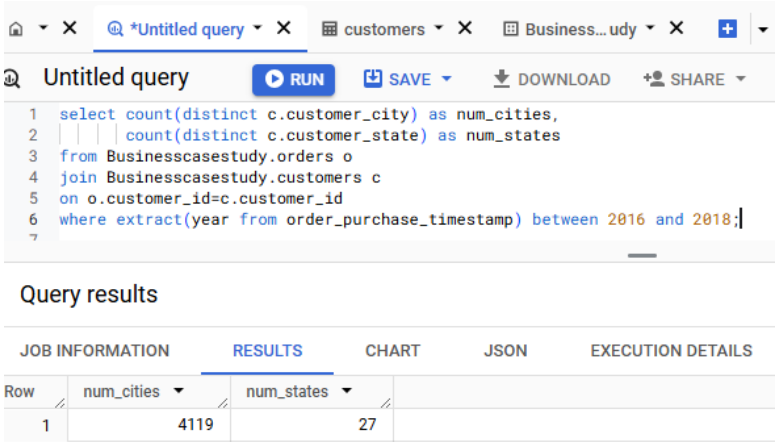
- It (Company/Client) is only asking the first purchase and last purchase timings.
- It is better to find time along with date i.e., on which dates the first purchase and last purchase are happened.
- We can also find customers who have done the first purchase and last purchase

3. Count the Cities & States of customers who ordered during the given period.

Query Ans:

```
select count(distinct c.customer_city) as num_cities,
       count(distinct c.customer_state) as num_states
from Businesscasestudy.orders o
join Businesscasestudy.customers c
on o.customer_id=c.customer_id
where extract(year from order_purchase_timestamp) between 2016 and 2018;
```

Output screenshot of 10 rows:



Insights:

- We can find the orders which are delivered to the customers for a particular zipcode.
- On which date the orders are delivered to the customers for a particular zipcode.
- Find the estimated date and time of the order for particular customer.
- At what time the maximum orders was placed by the customers who belongs to particular city.

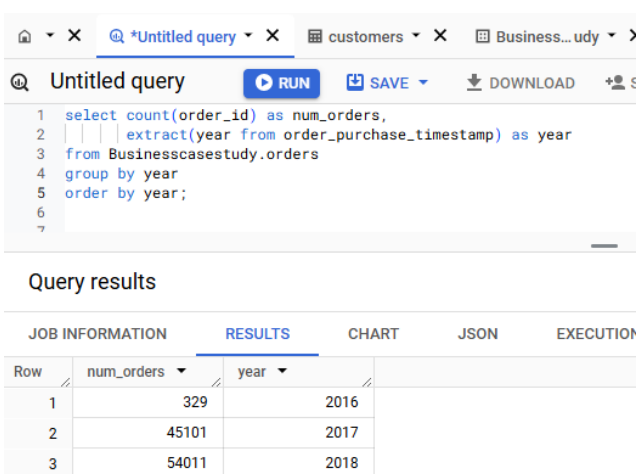
Q2: **In-depth Exploration:**

1. Is there a growing trend in the no. of orders placed over the past years?

Ans Query:

```
select count(order_id) as num_orders,
       extract(year from order_purchase_timestamp) as year
from Businesscasestudy.orders
group by year
order by year;
```

Output screenshot of 10 rows: Year by year sales are increasing



Insights:

- Year by year sales are increasing
- We can find the order status of all the orders done by customer.
- We can find the reason why this order is cancelled may be due to a long gap between order estimated delivery date and order delivery customer date of all the orders or it has strucked at carrier stage for a long time.

2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

Ans Query:

```
select extract(month from order_purchase_timestamp) as month,
              count(order_id) as month_wise_orders
from Businesscasestudy.orders
group by month
order by month;
```

Output screenshot of 10 rows:

Query results

JOB INFORMATION | RESULTS | CHART

Row	month	month_wise_orders
1	1	8069
2	2	8508
3	3	9893
4	4	9343
5	5	10573
6	6	9412
7	7	10318
8	8	10843
9	9	4305
10	10	4959
11	11	7544
12	12	5674

Insights:

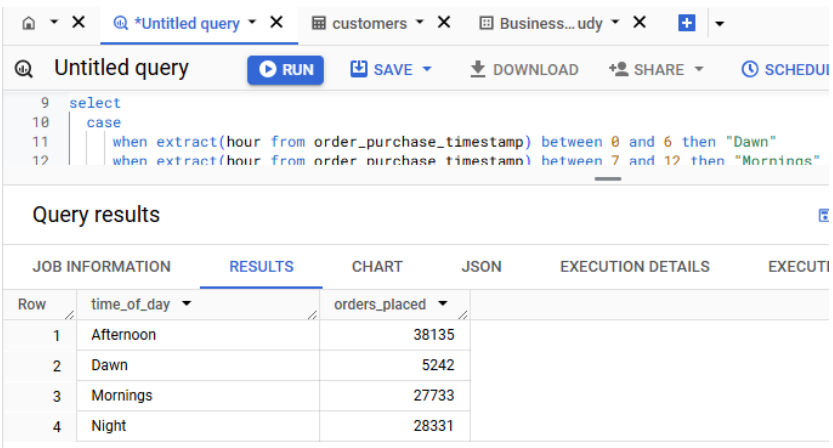
- In months 5,6,8 the sales are higher, to increase more sales by giving offers if it billing cost is above 11999 $.
- In months 9,10,12 the sales are low, we can increase the sales by giving more discounts on festival offers, special days
- We can see trend across each month between 2016 to 2018 years

3. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
    1. 0-6 hrs : Dawn
    2. 7-12 hrs : Mornings
    3. 13-18 hrs : Afternoon
    4. 19-23 hrs : Night

Ans Query:

```
select
   case
      when extract(hour from order_purchase_timestamp) between 0 and 6 then "Dawn"
      when extract(hour from order_purchase_timestamp) between 7 and 12 then "Mornings"
      when extract(hour from order_purchase_timestamp) between 13 and 18 then "Afternoon"
      else "Night"
   end as time_of_day,
   count(order_id) as orders_placed
from Businesscasestudy.orders
group by time_of_day
order by time_of_day,orders_placed desc;
```

Output screenshot of 10 rows: Brazilian customers mostly placed their orders in afternoon.



Row	time_of_day	orders_placed
1	Afternoon	38135
2	Dawn	5242
3	Mornings	27733
4	Night	28331

Insights:

- Brazilian customers mostly placed their orders in afternoon.
- In Dawn the order placed are very low. To increase the orders we can give offers on jumbo(combination of 2 or more) pack.
- To increase the orders on Dawn , we can give offer on mid journey pickups

Q3: **Evolution of E-commerce orders in the Brazil region:**

1. Get the month on month no. of orders placed in each state.

Ans Query:

```
select c.customer_state,
       extract(month from o.order_purchase_timestamp) as month,
       count(o.order_id) as month_wise_orders
from Businesscasestudy.orders o
join Businesscasestudy.customers c
on o.customer_id=c.customer_id
group by c.customer_state,month
order by c.customer_state,month;
```

Output screenshot of 10 rows:

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DE

Row	customer_state	month	month_wise_orders
1	AC	1	8
2	AC	2	6
3	AC	3	4
4	AC	4	9
5	AC	5	10
6	AC	6	7
7	AC	7	9
8	AC	8	7
9	AC	9	5
10	AC	10	6
11	AC	11	5
12	AC	12	5
13	AL	1	39
14	AL	2	39
15	AL	3	40
16	AL	4	51

Re

Insights:

- For example: The orders are less in months of march, sept, nov,desc for state 'AC' where we need to concentrate to increase the orders. We have to deep dive into it why the orders are less. We have to check whether the transportation/carrier station is properly functioning or not, whether our goods are properly send to customer without damages and without delay, is it a heavy material etc.

2. How are the customers distributed across all the states?

Ans Query:

```
select customer_state,
       count(distinct customer_unique_id) as customers_each_state
from Businesscasestudy.customers
group by customer_state;
```

Output screenshot of 10 rows:

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	customer_state	customers_each_sta
1	AC	77
2	AL	401
3	AM	143
4	AP	67
5	BA	3277
6	CE	1313
7	DF	2075
8	ES	1964
9	GO	1952
10	MA	726
11	MG	11259
12	MS	694
13	MT	876
14	PA	949
15	PB	519
16	PE	1609

Activate Windows
Go to Settings to activate W

Results per page: 50 1 – 27 of 27

Insights:

- We can find no. of customers for each citywise, statewise and their zipcodes in brazil.
- We can track the order using geolocation using zipcode,latitude,longitude

Q4: **Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.**

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only). You can use the "payment_value" column in the payments table to get the cost of orders.

Ans Query:

```
with cost as
(
   select extract(year from o.order_purchase_timestamp) as year,
          round(sum(p.payment_value),2) as cost_per_year
   from Businesscasestudy.orders o
   join Businesscasestudy.payments p
   on o.order_id=p.order_id
   where extract(year from o.order_purchase_timestamp) between 2017 and 2018
         and extract(month from o.order_purchase_timestamp) between 1 and 8
   group by year
),
minmax as
(
  select min(cost_per_year) as mini,
         max(cost_per_year) as maxi
  from cost
)
select cost.year,
       cost.cost_per_year,
       round(((maxi-mini)/mini)*100,2) as increase_percentage
from  minmax,cost;
```

Output screenshot of 10 rows:

JOB INFORMATION | RESULTS | CHART | JSON | EX

Row	year	cost_per_year	increase_percentage
1	2017	3669022.12	136.98
2	2018	8694733.84	136.98

Insights:

- The % of cost_per_year is increased in 2018 may be beause of recently introduced payment type(credit card).
- We can estimate the time taken for purchase till product delivered to customer.
- We can estimate whether the payment is in sequential(regularly) manner or not if it is EMI.

2. Calculate the Total & Average value of order price for each state.

Ans Query:

```
select c.customer_state,
       round(sum(ot.price),2) as Total_price,
       round(avg(ot.price),2) as Average_price
from Businesscasestudy.order_items ot
join Businesscasestudy.orders o on ot.order_id=o.order_id
join Businesscasestudy.customers c on o.customer_id=c.customer_id
group by c.customer_state
order by c.customer_state;
```

Output screenshot of 10 rows:

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	customer_state	Total_price	Average_price
1	AC	15982.95	173.73
2	AL	80314.81	180.89
3	AM	22356.84	135.5
4	AP	13474.3	164.32
5	BA	511349.99	134.6
6	CE	227254.71	153.76
7	DF	302603.94	125.77
8	ES	275037.31	121.91
9	GO	294591.95	126.27
10	MA	119648.22	145.2

Results per page: 50 1 – 27 of 27

Insights:

- We can estimate what are all the products ordered per order.
- We can derive the dimensions of the product.
- We can generate a report that who is going to sell these products

3. Calculate the Total & Average value of order freight for each state.

Ans Query:

```
select c.customer_state,
       round(sum(ot.freight_value),2) as Total_freight,
       round(avg(ot.freight_value),2) as Average_freight
from Businesscasestudy.order_items ot
join Businesscasestudy.orders o on ot.order_id=o.order_id
join Businesscasestudy.customers c on o.customer_id=c.customer_id
group by c.customer_state
order by c.customer_state;
```

Output screenshot of 10 rows:

Query results

SAVE RESULTS OPEN

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	customer_state	Total_freight	Average_freight
1	AC	3686.75	40.07
2	AL	15914.59	35.84
3	AM	5478.89	33.21
4	AP	2788.5	34.01
5	BA	100156.68	26.36
6	CE	48351.59	32.71
7	DF	50625.5	21.04
8	ES	49764.6	22.06
9	GO	53114.98	22.77
10	MA	31523.77	38.26

Results per page: 50 1 – 27 of 27

Activate Wind
Go to Settings to

Insights:

- If freight value is high then the distance b/w source to destination is large
- For example the average freight value is higher for state 'AC' means that state is far from source state
- Here we can introduce mode of transportation i.e., roadways,shipway,airway then it is easy to identify why freight value is high.

Q5: **Analysis based on sales, freight and delivery time.**

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
   Also, calculate the difference (in days) between the estimated & actual delivery date of an order.
   Do this in a single query.

   You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- **time_to_deliver** = order_delivered_customer_date - order_purchase_timestamp
- **diff_estimated_delivery** = order_delivered_customer_date - order_estimated_delivery_date

Ans Query:

```
select order_id,
       round(avg(date_diff(order_delivered_customer_date,order_purchase_timestamp,day)),2) as
          time_to_deliver,
       round(avg(date_diff(order_delivered_customer_date,order_estimated_delivery_date,day)),2) as
          diff_estimated_delivery
from Businesscasestudy.orders
where order_estimated_delivery_date is not null or order_delivered_customer_date is not null
      or order_purchase_timestamp is not null
group by order_id
order by order_id;
```

Output screenshot of 10 rows:

Row	order_id	time_to_deliver	diff_estimated_delive
1	00010242fe8c5a6d1ba2dd792...	7.0	-8.0
2	00018f77f2f0320c557190d7a1...	16.0	-2.0
3	000229ec398224ef6ca0657da...	7.0	-13.0
4	00024acbcdf0a6daa1e931b03...	6.0	-5.0
5	00042b26cf59d7ce69dfabb4e...	25.0	-15.0
6	00048cc3ae777c65dbb7d2a06...	6.0	-14.0
7	00054e8431b9d7675808bcb8...	8.0	-16.0
8	000576fe39319847cbb9d288c...	5.0	-15.0
9	0005a1a1728c9d785b8e2b08...	9.0	0.0
10	0005f50442cb953dcd1d21e1f...	2.0	-18.0

Results per page: 50 1 – 50 of 99441

Activate Win
Go to Settings to

Insights:

- We can infer from above result is if time to deliver is less than estimated delivery means the order is delivered within the estimated time for all states.
- By this we can know which order are delivered within estimated time and which orders are got delayed.
- We can find the reason for the delay of order whether it is strucked at carrier station or missed.

2. Find out the top 5 states with the highest & lowest average freight value.

Ans Query:

```
select * from
(
(select c.customer_state,
        round(avg(ot.freight_value),2) as average_freight
from Businesscasestudy.order_items ot
join Businesscasestudy.orders o on ot.order_id=o.order_id
join Businesscasestudy.customers c on o.customer_id=c.customer_id
group by c.customer_state
order by  average_freight desc
LIMIT 5
)
UNION ALL
(select c.customer_state,
       round(avg(ot.freight_value),2) as average_freight
from Businesscasestudy.order_items ot
join Businesscasestudy.orders o on ot.order_id=o.order_id
join Businesscasestudy.customers c on o.customer_id=c.customer_id
group by c.customer_state
order by  average_freight asc
LIMIT 5
)
)
order by average_freight desc;
```

Output screenshot of 10 rows:

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	customer_state	average_freight
1	RR	42.98
2	PB	42.72
3	RO	41.07
4	AC	40.07
5	PI	39.15
6	DF	21.04
7	RJ	20.96
8	MG	20.63
9	PR	20.53
10	SP	15.15

Results per page: 50 1 – 10 of 10

Insights:

- For example the average freight value is higher for state ‘RR’ means that state is far from source state
- For example the average freight value is low for state ‘SP’ means that state is nearer from source state. So the freight charges are also low.
- We can find the order reviews,rating,comments on each order.

3. Find out the top 5 states with the highest & lowest average delivery time.

Ans Query:

```
select customer_state, average_time_days from
(
(
select c.customer_state,
       round(avg(date_diff(o.order_delivered_customer_date,o.order_purchase_timestamp,day)),2)
as average_time_days
from Businesscasestudy.orders o
join Businesscasestudy.customers c
on o.customer_id=c.customer_id
where o.order_delivered_customer_date is not null or o.order_purchase_timestamp is not null
group by c.customer_state
order by  average_time_days desc
LIMIT 5
)
UNION ALL
(
select c.customer_state,
       round(avg(date_diff(o.order_delivered_customer_date,o.order_purchase_timestamp,day)),2)
as average_time_days
from Businesscasestudy.orders o
join Businesscasestudy.customers c
on o.customer_id=c.customer_id
where o.order_delivered_customer_date is not null or o.order_purchase_timestamp is not null
group by c.customer_state
order by  average_time_days asc
LIMIT 5
)
)
```

```
order by average_time_days desc;
```

Output screenshot of 10 rows:

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	customer_state	average_time_days
1	RR	28.98
2	AP	26.73
3	AM	25.99
4	AL	24.04
5	PA	23.32
6	SC	14.48
7	DF	12.51
8	MG	11.54
9	PR	11.53
10	SP	8.3

Results per page: 50 1 – 10 of 10

Insights:

- By this we can know how many days taken to reach the order to customer from purchase day to delivered day for each state.
- We can find the reason for the delay of order whether it is strucked at carrier station or missed.

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery. You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

Ans Query:

```
select c.customer_state,
       round(avg(date_diff(o.order_estimated_delivery_date,o.order_delivered_customer_date,day)),2) as
average_fast_days
from Businesscasestudy.orders o
join Businesscasestudy.customers c on o.customer_id=c.customer_id
where o.order_estimated_delivery_date is not null or o.order_delivered_customer_date is not null
group by c.customer_state
order by average_fast_days desc
LIMIT 5;
```

Output screenshot of 10 rows:

Query results

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTIC

Row	customer_state	average_fast_days
1	AC	19.76
2	RO	19.13
3	AP	18.73
4	AM	18.61
5	RR	16.41

Insights:

- If the difference b/w estimated delivery date and customer delivery date is less then we can say this is the fastest delivered order.
- If the difference b/w estimated delivery date and customer delivery date is high and it exceeds the estimated delivery date then that order is considered as delay.

Q6: **Analysis based on the payments:**

1. Find the month on month no. of orders placed using different payment types.

Ans Query:

```
select extract(month from o.order_purchase_timestamp) as month,
       p.payment_type,
       count(o.order_id) as num_orders
from Businesscasestudy.payments p
join Businesscasestudy.orders o
on p.order_id=o.order_id
group by payment_type,month
order by month;
```

Output screenshot of 10 rows:

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	month	payment_type	num_orders
1	1	voucher	477
2	1	credit_card	6103
3	1	UPI	1715
4	1	debit_card	118
5	2	voucher	424
6	2	credit_card	6609
7	2	UPI	1723
8	2	debit_card	82
9	3	voucher	591
10	3	credit_card	7707

Results per page: 50 1 – 50 of 50

Insights:

- We can able to know what are the payment modes.
- When we observe the data every month, no. of orders are more for the credit_card(EMI) payment type.
- We can give offer where no. of orders are less via debit_ card, like if you pay through credit card then you will get no-cost EMI (No interest imposed upto 1year) ,or some cash back since it is a new feature and customers will pay immediately.

2. Find the no. of orders placed on the basis of the payment installments that have been paid.

Ans Query:

```
select p.payment_installments,
       count(distinct o.order_id) as num_orders
from Businesscasestudy.payments p
join Businesscasestudy.orders o
on p.order_id=o.order_id
where p.payment_installments>0
group by payment_installments
order by payment_installments;
```

Output screenshot of 10 rows:

Query results

SAVE RESULTS

JOB INFORMATION | RESULTS | CHART | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	payment_installment	num_orders
1	1	49060
2	2	12389
3	3	10443
4	4	7088
5	5	5234
6	6	3916
7	7	1623
8	8	4253
9	9	644
10	10	5315

Results per page: 50 1 – 23 of 23

Insights:

- We can infer info.like no.orders who have done their first installments.
- How many payment installments has been done by which order and mode of payment